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TUT-6.

$$T(n) = aT\left(\frac{n}{b}\right) + f(n)$$

a) $T(n) = 3T(n/2) + n^2$

→

$a = 3 \quad b = 2 \quad f(n) = n^2$

$\log_b a = \log_2 3$

$n^{\log_2 3} < n^2$

So Case 3

$f(n) = \Omega(n^{p+\epsilon})$

$\Downarrow$

$T(n) = \Theta(f(n))$

hence

$T(n) = \Theta(n^2)$.

NOTES

b) $T(n) = 2T(n/2) + n$ $\quad T(1) = 1$

→ $a = 2 \quad b = 2 \quad f(n) = n$

$\log_b a = 1$

$n^{\log_b a} = n$

Use case 2

$T(n) = \Theta(n^{\log_b a} \log^{k+1} n)$

$$\boxed{T(n) = \Theta(n \log n)}$$

c) $T(n) = 3T(n/4) + \Theta(n^2)$

→ $a = 3 \quad b = 4 \quad f(n) = \Theta(n^2)$

$\log_4 3 \approx 0.79$

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$$n^{\log_b a} < n^2$$

Use case 3

$$T(n) = \Theta(f(n))$$

$$f(n) = \Omega(n^{\log_b a + \epsilon})$$

$$\boxed{T(n) = \Theta(n^2).}$$

d) $T(n) = 9T(n/3) + n$

→

$a = 9 \quad b = 3 \quad f(n) = n$

$\log_b a = \log_3 9 = 2$

$n^{\log_b a} > n$ ie $n^2 > n$

Use case 1

$$T(n) = \Theta(n^{\log_b a})$$

$$\boxed{T(n) = \Theta(n^2)}$$

e) $T(n) = 2T(\sqrt{n}) + \log(n)$

→ ~~a = 2~~ convert to master

$n = 2^m$ so $m = \log_2 n$

It becomes

$T(2^m) = 2T(2^{m/2}) + m.$

Sub as $P(m)$ so

$P(m) = 2P(m/2) + m$

Now

$a = 2 \quad b = 2 \quad f(n) = m$

$\log_b a = 1$

$n^{\log_b a} = n$

ie. Use case 2

$P(m) = \Theta(m \log m)$

NOTES

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Now reverse

$m = \log n$

$T(n) = P(\log n)$

$= \Theta(\log n . \log(\log n))$

$$\boxed{T(n) = \Theta(\log n . \log(\log n))}$$